
IT-SIMPLERA INSTITUTE

Cyber Security Analyst Internship program



INTERNSHIP WEEKLY REPORT

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Task 1: Five real-world cyberattacks

#	Cyberattack	Year	Organization	Attack Type	Description	Impact
1	WannaCry	2017	NHS, FedEx, Renault and others	Ransomware	WannaCry exploited the EternalBlue vulnerability in Microsoft Windows, encrypted victims' files, and demanded ransom payments in Bitcoin. It spread automatically across networks.	Over 230,000 computers in more than 150 countries were infected, causing billions of dollars in losses and disrupting healthcare services.
2	Equifax Data Breach	2017	Equifax	Data Breach	Attackers exploited an unpatched Apache Struts vulnerability and gained unauthorized access to customer databases containing sensitive personal information.	Personal information of approximately 147 million people was exposed, resulting in financial losses and legal penalties.
3	SolarWinds Attack	2020	SolarWinds	Supply Chain Attack	Attackers inserted malicious code into SolarWinds Orion software updates,	Numerous government agencies and major companies were compromised, leading to

					allowing unauthorized access to government agencies and private organizations.	large-scale cyber espionage.
4	Colonial Pipeline	2021	Colonial Pipeline	Ransomware	The DarkSide ransomware group encrypted company systems, forcing the shutdown of a major fuel pipeline in the United States.	Fuel shortages occurred across the U.S. East Coast, and the company paid millions of dollars in ransom.
5	Yahoo Data Breach	2013 - 2014	Yahoo	Data Breach	Cybercriminals gained unauthorized access to Yahoo's user database and stole billions of user account records.	Approximately 3 billion user accounts were compromised, damaging Yahoo's reputation and company value.

Task2

Understanding the CIA Triad

The CIA Triad is the foundation of information security. It consists of three core principles that help protect information and computer systems from cyber threats. These principles are Confidentiality, Integrity, and Availability.

CIA Principle	Description
Confidentiality	Ensures that sensitive information is accessible only to authorized users. Violations include unauthorized access, data theft, and information leakage. Typical security controls include

	encryption, passwords, multi-factor authentication (MFA), and access control.
Integrity	Ensures that data remains accurate, complete, and unaltered by unauthorized users. Violations include data modification, tampering, and unauthorized changes. Common controls include hashing, digital signatures, checksums, and version control.
Availability	Ensures that systems, networks, and data remain available to authorized users whenever needed. Violations include ransomware attacks, hardware failures, and Denial-of-Service (DoS/DDoS) attacks. Typical controls include backups, disaster recovery plans, redundant systems, and firewalls.

CIA Triad Mapping to Researched Cyberattacks

The table below shows which component(s) of the CIA Triad were affected by each researched cyberattack.

Cyberattack	Confidentiality	Integrity	Availability
WannaCry (2017)	Not directly	Not directly	Violated
Equifax Data Breach (2017)	Violated	Not directly	Not directly
SolarWinds Attack (2020)	Violated	Violated	Not directly
Colonial Pipeline (2021)	Not directly	Not directly	Violated
Yahoo Data Breach (2013–2014)	Violated	Not directly	Not directly

Task 3: Research Three Cybersecurity Tools

1. Wireshark

Purpose

Wireshark is a network protocol analyzer used to capture and inspect data packets traveling across a network.

Function

- Captures live network traffic.
- Displays packet details such as source and destination IP addresses.
- Helps identify suspicious or malicious network activity.
- Assists in troubleshooting network issues.

Use Case

A cybersecurity analyst uses Wireshark to detect unauthorized network connections, investigate malware communication, and diagnose network problems.

Advantages

- Free and open-source.
- Supports hundreds of network protocols.
- Easy packet filtering and analysis.

Limitations

- Requires networking knowledge.
- Large captures can be difficult to analyze.

2. Nmap (Network Mapper)

Purpose

Nmap is an open-source tool used to discover devices and services running on a network.

Function

- Detects active hosts.
- Scans open ports.
- Identifies operating systems.
- Finds running services and software versions.
- Detects security vulnerabilities.

Use Case

A security analyst performs regular network scans to identify open ports that attackers could exploit.

Advantages

- Fast and accurate.
- Supports multiple scan types.
- Widely used in penetration testing.

Limitations

- Can be detected by intrusion detection systems.
- Incorrect usage may trigger security alerts.

3. Metasploit Framework

Purpose

Metasploit is a penetration testing framework used to identify and verify system vulnerabilities.

Function

- Tests system security.
- Simulates cyberattacks.
- Validates vulnerabilities.
- Generates security reports.

Use Case

An ethical hacker uses Metasploit to determine whether a vulnerability can actually be exploited before recommending security improvements.

Advantages

- Large database of exploits.
- Automates penetration testing.
- Supports custom modules.

Limitations

- Should only be used with authorization.

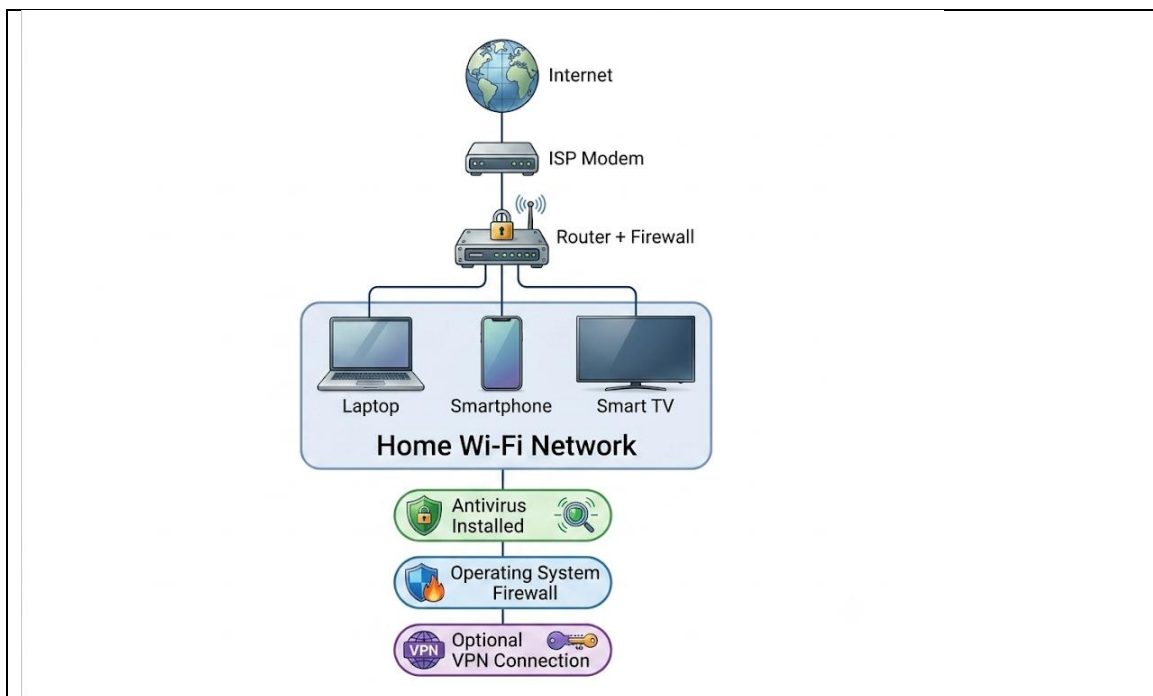
- Requires cybersecurity knowledge.

Comparison of Cybersecurity Tools

Tool	Purpose	Main Function	Used By
Wireshark	Network Monitoring	Packet Capture & Analysis	Network Security Analysts
Nmap	Network Scanning	Discover Hosts & Open Ports	Penetration Testers
Metasploit	Penetration Testing	Exploit Vulnerabilities	Ethical Hackers

Task 4: Secure Home Network Diagram

Diagram



Explanation

Internet

Provides access to websites and online services.

ISP Modem

Connects the home to the Internet Service Provider.

Router

Distributes the internet connection to all home devices.

Firewall

Blocks unauthorized access and filters incoming and outgoing traffic.

Antivirus

Protects devices from viruses, malware, ransomware, and spyware.

Devices

Examples include:

- Laptop
- Desktop Computer
- Smartphone
- Tablet
- Smart TV

VPN (Virtual Private Network)

Encrypts internet traffic and protects user privacy, especially on public Wi-Fi.

Task 5: What Does a Cybersecurity Analyst Do Every Day?**Introduction**

A cybersecurity analyst protects an organization's computer systems, networks, and sensitive information from cyber threats. Their primary responsibility is to monitor systems, detect attacks, and respond quickly to security incidents.

Daily Responsibilities

1. Monitor Network Activity

Cybersecurity analysts continuously monitor network traffic using security monitoring tools to identify suspicious activities and prevent cyberattacks.

2. Detect Security Threats

They investigate:

- Malware
- Ransomware
- Phishing attacks
- Unauthorized access
- Data breaches

3. Analyze Security Logs

They review logs generated by:

- Firewalls
- Servers
- Antivirus software
- Intrusion Detection Systems (IDS)

These logs help identify abnormal activities.

4. Respond to Security Incidents

When an attack occurs, analysts:

- Isolate infected systems.
- Remove malware.
- Recover lost data.
- Restore services.
- Investigate the root cause.

5. Perform Vulnerability Assessments

Analysts scan systems for weaknesses using tools like:

- Nmap
- Nessus
- OpenVAS

They recommend fixes before attackers can exploit vulnerabilities.

6. Install Security Updates

They ensure:

- Operating systems are updated.
- Applications are patched.
- Firmware is upgraded.

Regular updates reduce security risks.

7. Educate Employees

Cybersecurity analysts conduct awareness training on:

- Strong passwords
- Safe browsing
- Email security
- Phishing prevention
- Social engineering

8. Create Security Reports

Analysts prepare reports describing:

- Security incidents
- Vulnerabilities found
- Risks
- Recommendations
- Compliance status

Skills Required

- Networking
- Operating Systems
- Linux

- Windows Security
- Risk Assessment
- Incident Response
- Communication Skills
- Problem Solving

Common Tools Used

- Wireshark
- Nmap
- Metasploit
- Microsoft Defender
- Splunk
- Nessus

Importance of a Cybersecurity Analyst

Cybersecurity analysts help organizations:

- Protect confidential information.
- Prevent cyberattacks.
- Reduce financial losses.
- Maintain customer trust.
- Ensure business continuity.
- Meet legal and regulatory requirements.

Conclusion

Cybersecurity analysts play a vital role in protecting digital systems from evolving cyber threats. By monitoring networks, detecting vulnerabilities, responding to incidents, and educating users, they help ensure the confidentiality, integrity, and availability of organizational information. As cyberattacks become more sophisticated, the demand for skilled cybersecurity analysts continues to grow.

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